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Meta-heuristic search algorithms in truss optimization: Research on stability and complexity analyses

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ABSTRACT

Although they are among the most researched real world engineering design problems, it is encountered with significant problems in the optimization of structural truss bar problems (TPs). The main reasons for these problems are (i) the studies on the optimization of TPs being carried out with different experimental settings, (ii) the competitor algorithms used in the optimization process being insufficient, and (iii) the stability analysis and computational complexity information of the algorithms are not investigated. It is designed a simulation environment with defined standards and a benchmarking suite consisting of nine TPs of three different types in order to eliminate these problems. It is found optimum solutions for all problems in the benchmark suite and it is introduced feasible solutions for the first time. According to the statistical analysis results, among the seventy-seven competing algorithms, the best ten algorithms showed competitive performance in the optimization of TPs are LSHADE-EpSin, LSHADE-CnEpSin, SHADE, LSHADE, GSK, FDB-AGDE, MPA, LRFDB-COA and BES. According to the results of the stability analysis carried out on feasible solutions, it has been determined the algorithms with the best success rate for planar type; LSHADE, LSHADE-CnEpSin, LSHADE-EpSin and GSK with 100% success rate, LSHADE-CnEpSin, LSHADE-EpSin, LSHADE and SHADE for Space type with 100% success rate, and LSHADE-EpSin with 69% success rate for frequency constrained type. When the performances of the algorithms are evaluated regardless of the problem type, LSHADE-EpSin has been the most stable algorithm with an overall success rate of 90% on the TPs.

Click here for the simulation environment containing the source codes of the ten competitive algorithms and the TP benchmark suite proposed in the article: https://www.mathworks.com/matlabcentral/fileexchange/131668-structural-truss-problems-benchmark-suite.

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1. Introduction

Truss problems (TPs) are classical engineering problems in which truss system elements are dimensioned and/or positioned under constraints such as stress, displacement and frequency, generally aiming at minimum structural weight. It is aimed at using less material with the optimization realized. Thus, it is lowered the construction cost. In this way, it is ensured that it can be used more effectively the limited resources being decreased day by day. Structural truss system problems are quite preferred to test the performance of newly developed or modified heuristic algorithms on real-world problems [1–3]. These types of structures are the most preferred civil engineering benchmark problems in the field of optimization because not only they do not contain as complex design criteria as reinforced concrete structures and do not need to detail the reinforcements but also they create constraints and objective functions more easily, and minimize the local regulation differences in sizing the truss bars [4,5]. However, such structural design problems require structural analysis and/or modal analysis in the design processes. Therefore, heuristic algorithms in solving such problems that do not require differentiation of the objective function are much more preferred than gradient-based optimization methods. On the other hand, it is appropriate to state that heuristic algorithms require relatively more computational attempts because they are not gradient based and use random search procedures.

Meta-heuristic search (MHS) algorithms are more effective in optimization of structural truss bar problems compared to deterministic methods, thanks to their stochastic operators, swarm intelligence and parallel search capabilities [1,2,4,5]. On the other hand, there are great variations in the performance of meta-heuristics depending on the characteristics of the problem. The

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